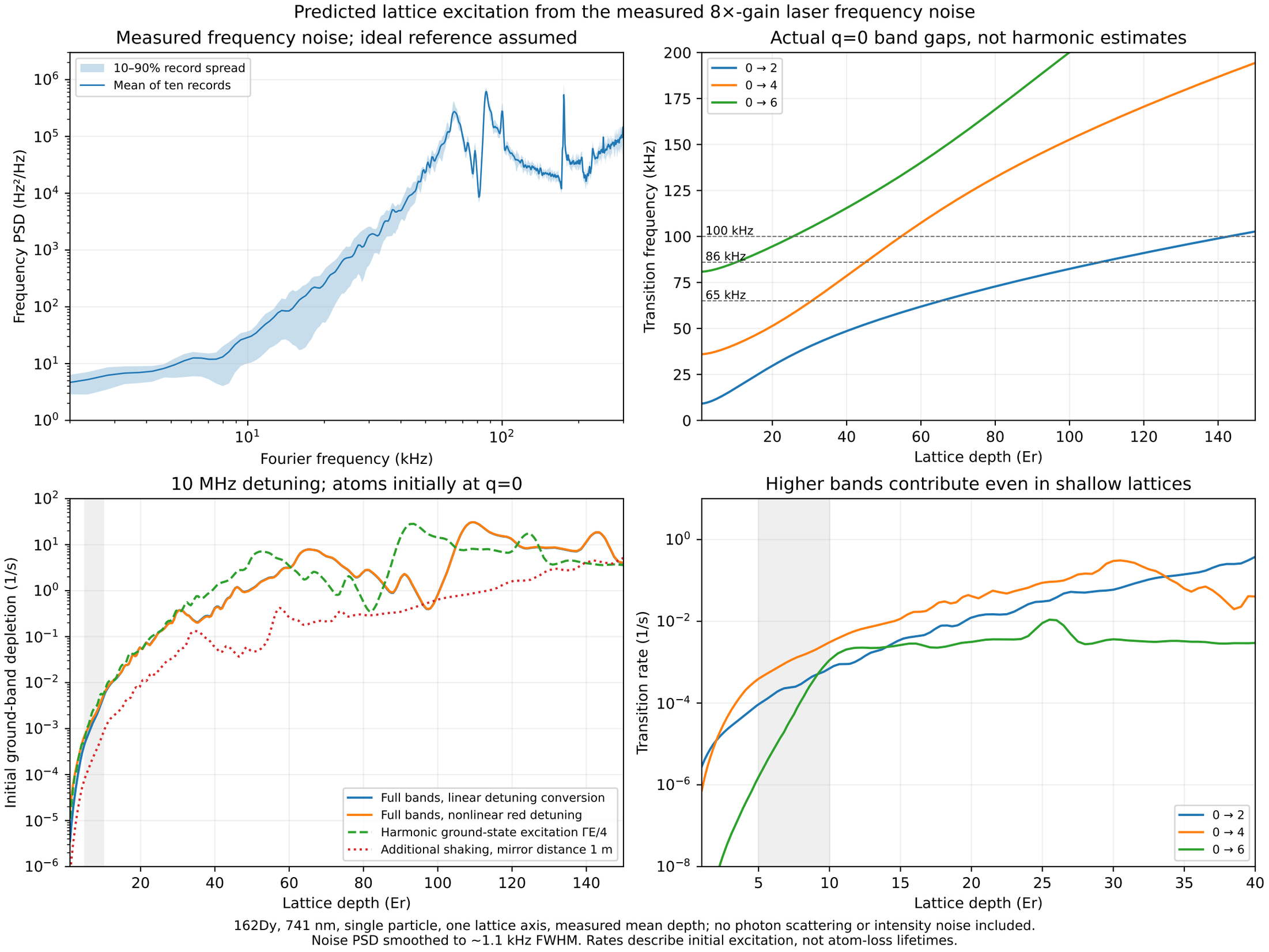

Predicted lattice excitation from the measured 8×-gain laser frequency noise
Measured frequency noise; ideal reference assumed
10–90% record spread
Mean of ten records
Frequency PSD (Hz²/Hz)
Fourier frequency (kHz)
Actual q=0 band gaps, not harmonic estimates
0 → 2
0 → 4
0 → 6
100 kHz
86 kHz
65 kHz
Transition frequency (kHz)
Lattice depth (Er)
10 MHz detuning; atoms initially at q=0
Full bands, linear detuning conversion
Full bands, nonlinear red detuning
Harmonic ground-state excitation ΓE/4
Additional shaking, mirror distance 1 m
Initial ground-band depletion (1/s)
Lattice depth (Er)
Higher bands contribute even in shallow lattices
0 → 2
0 → 4
0 → 6
Transition rate (1/s)
Lattice depth (Er)
162Dy, 741 nm, single particle, one lattice axis, measured mean depth; no photon scattering or intensity noise included.
Noise PSD smoothed to ~1.1 kHz FWHM. Rates describe initial excitation, not atom-loss lifetimes.